

ZAHRA MOJTAHEDIN

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Education

Sharif University Of Technology

Master of Science in Electrical Engineering (Digital Electronic Systems)

Major GPA:3.87/4.00

Tehran, Iran

Oct. 2024 – Jun. 2026

Sharif University Of Technology

Bachelor of Science in Electrical Engineering (Digital Systems)

Major GPA:3.56/4.00 (GPA of the first 6 semesters : 3.62/4.00)

Tehran, Iran

Oct. 2020 – Jun. 2024

Special school for elite student, Farzanegan High school

Mathematical and physics

Major GPA: 19.76/20.00

Tabriz, Iran

Sep. 2017 – Jun. 2020

Field of Interests

- Computer Architecture
- Embedded Systems
- Very Large Scale Integration
- ASIC/FPGA/System Design

Relevant Coursework

- ASIC/FPGA System Design & Lab
- Parallel Programming & Architecture
- Computer Structure & Lab
- Electronic Digital Circuits
- Embedded Systems Design
- Microprocessor System Design & Lab
- Advanced Computer Architecture
- Fault Tolerance Systems
- VLSI Systems Design
- Electronics I/II & Lab

Internship

Research Internship at USTC ([Wen Lab](#))

Mar. 2026 – Jun. 2026

- Developed neural recording hardware and embedded software for animal studies.
- Worked under Prof. Quan Wen on sensor design and neural data acquisition systems.

Summer Internship at [BIOSEN Group](#)

Jun. 2023 – Sep. 2023

- Programmed an STM32 microcontroller in C for an MCT cardiac monitoring device.
- Developed software for data acquisition, signal processing, and real-time transmission of heart activity.

Research Project

Architectural Optimization of Deep Neural Network Layers Using Approximation Theory

My master's thesis develops a theory-driven framework for designing neural network architectures using approximation theory. The project analyzes how depth, width, and connectivity affect approximation capability and proposes low-complexity design strategies that maintain accuracy, bridging the gap between theoretical limits and practical network construction.

Detection of Human Concentration Using Eyelid Movements and Neural Signals

My bachelor project at [ACL Lab](#) aims to develop a non-invasive and accurate method to monitor concentration employing advanced image processing and neural signal analysis techniques. An ESP32 board was used for the eye tracking component, which analyzes video images with a custom algorithm to detect eye coordinates. Data collected through eye tracking and EEG were analyzed to create a predictive model to estimate concentration levels.

Smart Wearables

I developed a multi-sensor wearable device capable of monitoring SpO₂, heart rate, ECG, and patient status. The system, built on an STM32 microcontroller with custom hardware design, uses LoRaWAN for secure long-range transmission and integrates algorithms for reliable real-time health monitoring.

Course Projects

Signal Processing on Zynq Board: This project aims to implement signal processing on a Zynq board, including scrambler, descrambler, and FFT on digital signals, with results displayed on an HDMI screen. We will generate different signal types (triangular, square, sinusoidal) and process them in the FPGA system.

SchmittWave Function Generator: The goal of this project is to design a 10V function generator using a Schmitt trigger circuit, capable of producing square, triangular and sinusoidal signals from 100Hz to 100KHz. The generator must create 21V square and triangular wave amplitudes, with less than 1.5% THD for the sinusoidal wave.

555 Timer Circuit Design: This project focuses on monostable and astable multivibrators, Schmitt triggers, and PWM. I use Proteus to create and test these circuits: a monostable circuit for single pulses, an astable circuit for continuous square waves, a PWM circuit for pulse modulation, and a Schmitt trigger for signal cleaning.

Implementation of four-issue MIPS processor:

For this project, I develop a four-issue MIPS processor by first implementing an execution processor "in order" and optimizing branch prediction mechanisms. Then, I enhance the processor for "Out-of-Order" execution, focusing on managing instruction dependencies and resource allocation.

Bipolar Amplifier Design: This project aims to design and simulate a circuit with a minimum gain of 76 dB, a maximum power consumption of 15 mW, and an output swing of at least 5.5 V using NPN transistors with parameters ($\beta = 100$, $V_a = 100\text{V}$, $I_s = 2\text{fA}$) and PNP transistors with parameters ($\beta = 100$, $V_a = 30\text{V}$, $I_s = 4\text{fA}$).

OpAmp Design: The objective of this project is to design a CMOS operational amplifier with specified parameters in $0.18\mu\text{m}$ technology. This design involves maximizing critical characteristics such as gain and CMRR while minimizing power consumption.

NAND Logic Reaction Timer: The purpose of this project is to design a reaction timer using NAND gates. This device includes LEDs and push buttons, where participants compete to press their button after a signal.

Sensing & Measurement: This project analyzes muscle activity during evoked contractions, focusing on activation, coactivation of antagonist muscles, and the link between muscle activity and force. It measures nerve conduction velocity through electrical stimulation to identify nerve and muscle disorders, using tools like LabChart for analysis.

Embedded Systems Design: This project develops an IoT-based embedded platform for real-time event detection and environmental monitoring. Integrates a Linux-based server with an ESP32 microcontroller for sensing, data transmission, and system control. The system detects audio events using neural network models, manages data through a database and web interface, and communicates via the MQTT protocol. It demonstrates modular embedded design principles and provides a reliable framework for smart monitoring applications.

Awards & Honors

- Acceptance of brilliant master's talent and NODET's quota in Electrical Engineering (Digital Systems) at SUT
- Scoring 101/120 in TOEFL test (Reading:30 , Listening:23 , Speaking:20 , Writing:28)
- Top %20 in B.Sc Electrical Engineering class at Sharif University of Technology
- Ranked first among-st the female students of the Digital Systems and Biomedical Engendering branch
- Ranked among-st the top %0.5 in the National University Entrance Exam

Work Experience

Laboratory & Teaching Assistant at Sharif University of Technology

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| • Computer Architecture | • Logic Circuits and Digital Systems |
| • Sensing and Measuring | • Electronics 2 |
| • Electronics 1 | • Microprocessor Laboratory |
| • Parallel Programming | • Pulse Technique |
| • Advanced Computer Structure | • Electronics Laboratory |

Skills

Languages

- English (Professional working proficiency)
- Persian (Native)
- Turkish (Advanced)
- Azerbaijani (Native)

Software

- Proteus
- ADS
- Arduino
- STM32CubeMX
- PSpice
- LTspice
- Altium Designer
- Vivado
- Quartus
- Linux
- Cadence
- Microsoft Office

Programming Language

- MIPS Assembly
- VHDL/Verilog
- Python
- MATLAB
- Java
- C/C++
- Laravel
- L^AT_EX

Volunteer Experience

- Event Graphics & Photographer Head | ReACT 2022, 2023, 2024
- Event Coordinator | Iran Workshop on Communication and Information Theory 2022
- Event Graphics Head | Emeet 2022
- Head Graphic Designer | Event of Life in Sharif 2021

Hobbies

- Volleyball Player of EE Department Women’s Team
- Graphic Designer
- Photographer
- Painter